

# Galvo Fiber Laser Serialization Marker

Authoritative Model Technical Manual & Diagnostic Reference | VAI-MAN-LM-001

**NOTICE: SYNTHETIC PROTOTYPE TECHNICAL MANUAL**

This document is artificially generated for the VectorAI demonstration and software development.

The specifications, thresholds, service-life values, maintenance intervals, and operating parameters are synthetic and must not be used for real industrial equipment operation or maintenance.

## 1. Machine Overview & Manufacturing Context

|                   |                                        |                        |                            |
|-------------------|----------------------------------------|------------------------|----------------------------|
| Equipment Model:  | Galvo Fiber Laser Serialization Marker | Document ID:           | VAI-MAN-LM-001             |
| Machine Type Key: | laser-marking                          | Cleanroom Bay / Stage: | Bay 5C: Laser Marking Cell |
| Document Version: | 1.0                                    | Release Date:          | 2026-08-27                 |

Pulsed 30W fiber laser marking and serialization system for inscribing high-contrast 2D DataMatrix barcodes, human-readable alphanumeric text, and lot codes onto molded package surfaces.

**Process Flow:** A 1064nm pulsed MOPA fiber laser delivers focused beam pulses through a dual-axis galvanometer scanner head and telecentric F-theta lens, vaporizing top epoxy resin monolayers without compromising package internal integrity.

## 2. Major Subsystems & Critical Components

| Subsystem / Component   | Primary Function                                                      | Key Parameters                                                    | Degradation Indicators                                   |
|-------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------------------|
| MOPA Fiber Laser Source | Emits 30W pulsed 1064nm laser beam.                                   | Average power (W), pulse frequency (kHz), diode pump current (A). | Laser power degradation (<26W), pulse shape instability. |
| Galvo Scanner Mirrors   | Steers laser beam in X/Y axes at velocities up to 8,000 mm/s.         | Galvo motor temp (°C), position feedback error (μrad).            | Galvo temp rise (>35°C), barcode edge distortion.        |
| F-Theta Scan Lens       | Focuses laser beam into flat focal plane across entire package strip. | Focal length (mm), optical transmission (%).                      | Epoxy fume residue on lens, beam spot widening.          |

### 3. Sensor Telemetry & Threshold Matrix

Threshold boundaries configured for real-time telemetry monitoring and automatic anomaly triggers:

| Sensor Name & ID                 | Unit | Normal Range | Warning Range | Critical Range | Direction       |
|----------------------------------|------|--------------|---------------|----------------|-----------------|
| Fiber Laser Power<br>laser_power | W    | 28 – 32      | 22 – 28       | 0 – 22         | Lower is Worse  |
| Galvo Head Temp<br>galvo_temp    | °C   | 24 – 32      | 32 – 42       | 42 – 60        | Higher is Worse |

### 4. Deterministic RUL Model (Weighted Degradation)

**Base Useful Life:** 3,500 operating hours. **Model:**  $RUL = BaseLife * (1 - (0.50 * wear\_laser\_power + 0.30 * wear\_galvo\_temp + 0.20 * wear\_exhaust))$

| Parameter        | Sensor ID    | Unit | Weight      | Healthy Limit | Critical Limit | Direction       |
|------------------|--------------|------|-------------|---------------|----------------|-----------------|
| Laser Power      | laser_power  | W    | 0.50        | 30            | 20             | LOWER IS WORSE  |
| Galvo Temp       | galvo_temp   | °C   | 0.30        | 28            | 45             | HIGHER IS WORSE |
| Exhaust Flow     | exhaust_flow | m/s  | 0.20        | 12            | 5              | LOWER IS WORSE  |
| TOTAL WEIGHT SUM | -            | -    | 1.00 (100%) | -             | -              | VERIFIED OK     |

5. Troubleshooting & Anomaly Symptoms Matrix

| Symptom & ID                                              | Severity | Related Sensors              | Possible Root Causes                                                                                                                    | Recommended Corrective Action                          |
|-----------------------------------------------------------|----------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| 2D DataMatrix Barcode Readability Grade < B<br>SYM-LM-001 | HIGH     | laser_power,<br>exhaust_flow | <ul style="list-style-type: none"><li>• F-theta lens smoke residue</li><li>• Laser power drop</li><li>• Exhaust filter loaded</li></ul> | Wipe scan lens and replace fume extraction pre-filter. |

6. Authoritative Diagnostic Knowledge Base (Failure Scenarios)

|                                                                                 |                                                                                                           |
|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Scenario 1: Planned Optical Window Cleaning & F-Theta Calibration (SCEN-LM-001) | Severity: LOW                                                                                             |
| Telemetry Sensor Pattern:                                                       | laser_power == 0 W (Machine offline)                                                                      |
| Possible Root Causes:                                                           | Scheduled 500-hour preventive maintenance interval                                                        |
| Recommended Corrective Action:                                                  | Complete optical lens wipe and run 2D DataMatrix verification target lot.                                 |
| Verification & Recovery Steps:                                                  | Verify laser power with thermal power meter (> 29.5W) > Scan verification target sheet (Grade A verified) |